

# **Software Requirements Specification**

## **for**

# **Hyperion Mobile Support Application**

**Version 1.1**

**Prepared by**

**PFM-SD (Applications Systems Unit )**

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# Table of Contents

|                                                    |    |
|----------------------------------------------------|----|
| Table of Contents.....                             | 2  |
| List of Abbreviations.....                         | 4  |
| Requirement ID Prefix Key .....                    | 5  |
| Functional Requirement Prefixes (FR-) .....        | 5  |
| Non-Functional Requirement Prefixes (NFR-).....    | 5  |
| 1. Introduction .....                              | 6  |
| 1.1 Purpose .....                                  | 6  |
| 1.2 Scope .....                                    | 6  |
| 1.3 References .....                               | 6  |
| 1.4 Overview.....                                  | 6  |
| 2. Overall Description .....                       | 7  |
| 2.1 Product Perspective.....                       | 7  |
| 2.2 Product Functions (Summary).....               | 7  |
| 2.3 User Classes and Characteristics .....         | 7  |
| 2.4 Operating Environment.....                     | 8  |
| 2.5 Design and Implementation Constraints.....     | 8  |
| 2.6 Assumptions and Dependencies .....             | 8  |
| 3. System Features (Functional Requirements) ..... | 9  |
| 3.1 User Authentication and Role-Based Access..... | 9  |
| 3.2 Issue Logging and Ticket Management.....       | 9  |
| 3.3 Notifications and Alerts .....                 | 10 |
| 3.4 Knowledge Base and Self-Service .....          | 10 |
| 3.5 Offline and Low-Bandwidth Support.....         | 11 |
| 3.6 Support Dashboard and Reporting.....           | 11 |
| 3.7 Security and Audit Trail .....                 | 11 |
| 4. External Interface Requirements.....            | 12 |
| 4.1 User Interfaces .....                          | 12 |
| 4.2 Hardware Interfaces .....                      | 12 |
| 4.3 Software Interfaces.....                       | 12 |
| 4.4 Communications Interfaces.....                 | 12 |
| 5. Non-Functional Requirements .....               | 13 |
| 6. Data Requirements.....                          | 14 |
| 7. Other Requirements.....                         | 15 |
| 7.1 Legal and Compliance.....                      | 15 |
| 7.2 Localization .....                             | 15 |
| 8. Use Case Diagram.....                           | 16 |
| Appendix A: Requirements Traceability Matrix ..... | 18 |

|                                                                                  |                                     |
|----------------------------------------------------------------------------------|-------------------------------------|
| <b>Appendix B: Open Assumptions and Items for Stakeholder Confirmation .....</b> | <b>19</b>                           |
| <b>Appendix C: UI Screens (Design Reference) .....</b>                           | <b>Error! Bookmark not defined.</b> |

## List of Abbreviations

The following abbreviations are used throughout this document.

| Abbreviation | Meaning                                       |
|--------------|-----------------------------------------------|
| MDA          | Ministry, Department, or Agency               |
| MMDA         | Metropolitan, Municipal, or District Assembly |
| PFM          | Public Financial Management                   |
| FCM          | Firebase Cloud Messaging                      |
| SRS          | Software Requirements Specification           |
| FR           | Functional Requirement                        |
| NFR          | Non-Functional Requirement                    |
| RBAC         | Role-Based Access Control                     |
| KB           | Knowledge Base                                |

## Requirement ID Prefix Key

Functional (FR-) and non-functional (NFR-) requirements in Sections 3 and 5 are identified by a feature-area prefix followed by a sequence number (e.g., FR-TCK-05). This key decodes each prefix; full requirement wording appears in the corresponding table.

### Functional Requirement Prefixes (FR-)

| Prefix | Feature Area                              | Section |
|--------|-------------------------------------------|---------|
| AUTH   | User Authentication and Role-Based Access | 3.1     |
| TCK    | Issue Logging and Ticket Management       | 3.2     |
| NOT    | Notifications and Alerts                  | 3.3     |
| KB     | Knowledge Base and Self-Service           | 3.4     |
| OFF    | Offline and Low-Bandwidth Support         | 3.5     |
| DSH    | Support Dashboard and Reporting           | 3.6     |
| SEC    | Security and Audit Trail                  | 3.7     |

### Non-Functional Requirement Prefixes (NFR-)

| Prefix | Category           | Section |
|--------|--------------------|---------|
| PERF   | Performance        | 5       |
| SEC    | Security           | 5       |
| REL    | Reliability        | 5       |
| AVAIL  | Availability       | 5       |
| USE    | Usability          | 5       |
| MAINT  | Maintainability    | 5       |
| PORT   | Portability        | 5       |
| SCAL   | Scalability        | 5       |
| OFF    | Offline capability | 5       |

# 1. Introduction

## 1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Hyperion Mobile Support Application, a mobile app intended to support Oracle Hyperion users across Ministries, Departments and Agencies (MDAs) and Metropolitan, Municipal and District Assemblies (MMDAs). It is intended for use by developers, reviewers, and stakeholders to agree on scope before implementation begins.

## 1.2 Scope

The application will provide issue logging, ticket tracking and escalation, push notifications, a searchable knowledge base, an administrative dashboard, offline and low-bandwidth support, and role-based, audited access control. It complements and does not replace the existing Oracle Hyperion platform. The initial release targets Android and iOS mobile devices, with a Firebase-backed serverless architecture as the reference implementation.

Out of scope for this version: direct integration with Oracle Hyperion / Essbase APIs, automated resolution of budget data errors, and a public-facing (non-authenticated) version of the app.

## 1.3 References

- Ministry of Finance Concept Note: “Strengthening Oracle Hyperion Support Services and Developing a Mobile Support Application for MDAs and MMDAs” (source document for this SRS).
- Rapid Implementation Roadmap & Sprint Execution Plan: Hyperion Mobile Support Application (companion execution roadmap document).

## 1.4 Overview

Section 2 describes the product at a high level. Section 3 specifies functional requirements by feature area. Section 4 specifies external interfaces.

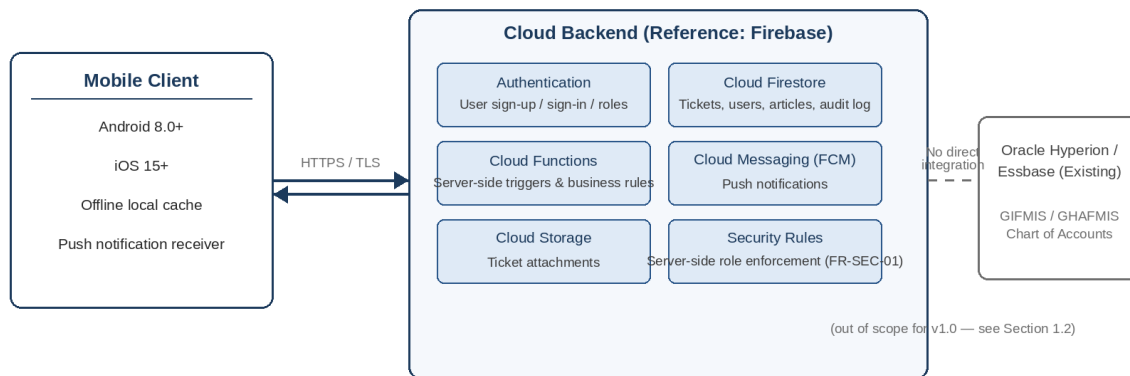
Section 5 specifies non-functional requirements. Section 6 describes data requirements. Section 7 lists other requirements. Appendices provide a traceability matrix and open assumptions.

## 2. Overall Description

### 2.1 Product Perspective

The application is a new, self-contained mobile client that integrates with a cloud backend (reference implementation: Firebase Authentication, Cloud Firestore, Cloud Functions, Cloud Messaging, and Cloud Storage). It does not modify Oracle Hyperion itself; it exists to structure and accelerate the human support process around it.

**Figure 2.1 — Hyperion Mobile Support Application: Reference Architecture**



The mobile client communicates only with the Firebase-backed cloud services over HTTPS/TLS. The application does not modify Oracle Hyperion; it structures and accelerates the human support process that surrounds it.

*Figure 2.1 — Reference architecture: mobile client, Firebase-backed cloud services, and the existing Oracle Hyperion platform (no direct integration in this version).*

### 2.2 Product Functions (Summary)

- Register and authenticate users with role-based access
- Log, categorize, and attach evidence to support issues
- Track, assign, and escalate tickets through defined statuses
- Notify users of ticket and system events
- Provide a searchable self-service knowledge base
- Operate usefully with intermittent or no connectivity
- Give management a dashboard view of support demand and performance
- Maintain an auditable record of key actions

### 2.3 User Classes and Characteristics

| User Class          | Description                                                            | Technical Proficiency                     |
|---------------------|------------------------------------------------------------------------|-------------------------------------------|
| MDA / MMDA User     | Budget officers submitting and tracking their own support tickets.     | Basic smartphone literacy assumed.        |
| Support Coordinator | Triages, assigns, and follows up on tickets across institutions.       | Moderate; trained on the support process. |
| Functional Lead     | Handles budget process, workflow, and reporting escalations.           | Moderate to high domain expertise.        |
| Technical Lead      | Handles metadata, security, Essbase, and technical escalations.        | High technical expertise.                 |
| Administrator       | Manages users, roles, knowledge base content, and views the audit log. | High system familiarity.                  |

## 2.4 Operating Environment

Client: Android 8.0+, iOS 15+ smartphones and tablets. Backend: cloud-hosted serverless services (reference: Firebase, region-configurable). Network: designed to tolerate intermittent 2G/3G/4G connectivity typical of field and district-level use.

The web version will be implemented using Flutter, leveraging the same reference database (Firebase) used across the platform for consistency and centralized data management.

## 2.5 Design and Implementation Constraints

- Must not require modification to the existing Oracle Hyperion deployment.
- Must support offline operation for core ticket-logging functions (see FR-OFF-01 to FR-OFF-05).
- Data residency and hosting must be confirmed with the Ministry's IT/security policy before any production pilot (see Section 7).
- Initial release is mobile-only; no requirement for a web administrative console in this version.

## 2.6 Assumptions and Dependencies

- Users have access to a smartphone and at least intermittent internet connectivity.
- The Ministry will designate institutional focal persons and an escalation structure, per the source concept note.
- Push notification delivery depends on the mobile OS vendor's notification service (FCM/APNs) being reachable.

### 3. System Features (Functional Requirements)

#### 3.1 User Authentication and Role-Based Access

| ID         | Requirement                                                                                                                                      | Priority |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-AUTH-01 | The system shall allow a user to register with email, password, institution name, and requested role.                                            | High     |
| FR-AUTH-02 | The system shall allow a registered user to sign in with email and password.                                                                     | High     |
| FR-AUTH-03 | The system shall assign each user exactly one role: MDA User, MMDA User, Support Coordinator, Functional Lead, Technical Lead, or Administrator. | High     |
| FR-AUTH-04 | The system shall restrict access to screens and actions based on the authenticated user's role.                                                  | High     |
| FR-AUTH-05 | The system shall allow a user to sign out, invalidating the local session.                                                                       | Medium   |
| FR-AUTH-06 | The system shall allow an Administrator to change a user's role or deactivate a user account.                                                    | Medium   |
| FR-AUTH-07 | The system shall enforce password rules (minimum length 8, at least one number) at registration.                                                 | Medium   |

#### 3.2 Issue Logging and Ticket Management

| ID        | Requirement                                                                                                                                                                 | Priority |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-TCK-01 | The system shall allow an authenticated user to submit a support ticket with category, description, and an optional attachment (image or document).                         | High     |
| FR-TCK-02 | The system shall classify tickets: Access, Workflow, Budget Forms, Reports, Smart View, Business Rules, Metadata, Data Validation, System Performance, and General Enquiry. | High     |
| FR-TCK-03 | The system shall auto-generate a unique, human-readable ticket reference (e.g., TCK-XXXXXXX) upon submission.                                                               | High     |
| FR-TCK-04 | The system shall record the submitting user's institution and role on the ticket.                                                                                           | High     |
| FR-TCK-05 | The system shall allow a user to view a list of their own submitted tickets, filterable by status and category.                                                             | High     |
| FR-TCK-06 | The system shall allow Support Coordinators and above to view all tickets across institutions.                                                                              | High     |
| FR-TCK-07 | The system shall allow Support Coordinators and above to assign a ticket to a specific officer.                                                                             | High     |
| FR-TCK-08 | The system shall allow authorized users to update ticket status: Open, In Progress, Escalated, Resolved, Reopened, Closed.                                                  | High     |

| ID        | Requirement                                                                                                                                                 | Priority |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-TCK-09 | The system shall allow authorized users to escalate a ticket to a Functional Lead or Technical Lead.                                                        | High     |
| FR-TCK-10 | The system shall allow authorized users to record resolution notes when closing a ticket.                                                                   | Medium   |
| FR-TCK-11 | The system shall allow a ticket creator to reopen a ticket previously marked Resolved within 14 days of resolution.                                         | Medium   |
| FR-TCK-12 | The system shall record a priority level (Low, Medium, High, Critical) on each ticket, settable at creation and editable by Support Coordinators and above. | Medium   |

### 3.3 Notifications and Alerts

| ID        | Requirement                                                                                                                         | Priority |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-NOT-01 | The system shall send a push notification to the ticket creator when the ticket's status changes.                                   | High     |
| FR-NOT-02 | The system shall send a push notification to a support officer when a ticket is assigned to them.                                   | High     |
| FR-NOT-03 | The system shall send a push notification when a ticket is escalated.                                                               | Medium   |
| FR-NOT-04 | The system shall allow Administrators to broadcast a notification to all users (e.g., planned maintenance, budget cycle deadlines). | Medium   |
| FR-NOT-05 | The system shall display in-app notifications when the app is in the foreground.                                                    | Medium   |
| FR-NOT-06 | The system shall allow a user to view a history of their notifications within the app.                                              | Low      |

### 3.4 Knowledge Base and Self-Service

| ID       | Requirement                                                                                                       | Priority |
|----------|-------------------------------------------------------------------------------------------------------------------|----------|
| FR-KB-01 | The system shall provide a searchable list of knowledge base articles (FAQs, guides, budget-cycle support notes). | Medium   |
| FR-KB-02 | The system shall allow filtering knowledge base articles by category.                                             | Medium   |
| FR-KB-03 | The system shall allow Administrators to create, edit, and remove knowledge base articles.                        | Medium   |
| FR-KB-04 | The system shall allow a user to view an article's full content offline once it has been loaded at least once.    | Low      |

### 3.5 Offline and Low-Bandwidth Support

| ID        | Requirement                                                                                                      | Priority |
|-----------|------------------------------------------------------------------------------------------------------------------|----------|
| FR-OFF-01 | The system shall detect loss of network connectivity and indicate offline status to the user.                    | High     |
| FR-OFF-02 | The system shall allow a user to compose and queue a new ticket while offline.                                   | High     |
| FR-OFF-03 | The system shall automatically submit queued tickets once connectivity is restored.                              | High     |
| FR-OFF-04 | The system shall cache the most recently viewed tickets and knowledge base articles locally for offline viewing. | Medium   |
| FR-OFF-05 | The system shall show a clear "pending sync" indicator on any ticket created while offline.                      | Medium   |

### 3.6 Support Dashboard and Reporting

| ID        | Requirement                                                                                                                  | Priority |
|-----------|------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-DSH-01 | The system shall display ticket volume by category and institution to Functional Leads, Technical Leads, and Administrators. | Medium   |
| FR-DSH-02 | The system shall display average resolution time over a selectable date range.                                               | Medium   |
| FR-DSH-03 | The system shall display counts of open, in-progress, escalated, and resolved tickets.                                       | Medium   |
| FR-DSH-04 | The system shall highlight recurring issue categories (top 5 by volume) over the last 30 days.                               | Low      |

### 3.7 Security and Audit Trail

| ID        | Requirement                                                                                                                      | Priority |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-SEC-01 | The system shall enforce role-based read/write access at the data layer (server-side security rules), not only in the client UI. | High     |
| FR-SEC-02 | The system shall record an audit entry (user, action, timestamp) for every ticket status change and role change.                 | High     |
| FR-SEC-03 | The system shall allow only Administrators to view the audit log.                                                                | Medium   |
| FR-SEC-04 | The system shall encrypt data in transit using TLS for all network communication.                                                | High     |
| FR-SEC-05 | The system shall time out an inactive session after 30 minutes and require re-authentication.                                    | Low      |

## 4. External Interface Requirements

### 4.1 User Interfaces

The application shall provide a mobile-native interface (Material Design on Android, Human Interface Guidelines conventions on iOS) with screens for: sign-in/registration, ticket list, new-ticket form, ticket detail, knowledge base, notifications, and (for authorized roles) an admin dashboard.

### 4.2 Hardware Interfaces

The application shall use the device camera and/or photo library for attachment capture, and local storage for offline caching. No specialized hardware is required.

### 4.3 Software Interfaces

| Interface                                                  | Purpose                                                            |
|------------------------------------------------------------|--------------------------------------------------------------------|
| Authentication provider (e.g., Firebase Authentication)    | User sign-up, sign-in, session management                          |
| Document database (e.g., Cloud Firestore)                  | Ticket, user, article, and audit data storage                      |
| Push notification service (e.g., Firebase Cloud Messaging) | Delivery of ticket and system notifications                        |
| Object storage (e.g., Cloud Storage)                       | Ticket attachment storage                                          |
| Serverless functions (e.g., Cloud Functions)               | Server-side triggers, e.g., notification dispatch on status change |

### 4.4 Communications Interfaces

All client-server communication shall use HTTPS/TLS. The application shall detect and gracefully handle loss of connectivity per Section 3.5.

## 5. Non-Functional Requirements

| ID           | Category           | Requirement                                                                                                                               |
|--------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-PERF-01  | Performance        | The ticket list shall render within 2 seconds on a standard 4G connection for up to 500 tickets.                                          |
| NFR-PERF-02  | Performance        | Ticket creation (online) shall confirm submission to the user within 3 seconds under normal network conditions.                           |
| NFR-SEC-01   | Security           | All user credentials shall be handled exclusively by the authentication provider; the app shall never store raw passwords.                |
| NFR-SEC-02   | Security           | File attachments shall be scanned for type/size limits (max 10 MB, images and PDFs only) before upload.                                   |
| NFR-REL-01   | Reliability        | The app shall not lose a locally queued offline ticket even if the app is closed before connectivity returns.                             |
| NFR-AVAIL-01 | Availability       | Backend services shall target 99.5% monthly uptime, consistent with the hosting provider's SLA.                                           |
| NFR-USE-01   | Usability          | A first-time user shall be able to submit a ticket without training, in 5 steps or fewer.                                                 |
| NFR-USE-02   | Usability          | The app shall support both English text and low-literacy-friendly iconography for common actions.                                         |
| NFR-MAINT-01 | Maintainability    | The codebase shall follow a feature-first module structure so that individual features can be modified without cross-feature regressions. |
| NFR-PORT-01  | Portability        | The app shall run on Android 8.0+ and iOS 15+.                                                                                            |
| NFR-SCAL-01  | Scalability        | The data model shall support all MMDAs and MDAs and 1 million tickets per year without redesign.                                          |
| NFR-OFF-01   | Offline capability | Core ticket-creation and knowledge-base-viewing functions shall remain usable with zero connectivity.                                     |

## 6. Data Requirements

The following entities represent the core data model. The exact schema is defined in the implementation guide; this table establishes the requirement-level data scope.

| Entity       | Key Fields                                                                                                                                        | Notes                                                            |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| User         | uid, email, role, institution, fcmToken                                                                                                           | One document per authenticated user; role drives access control. |
| Ticket       | id, reference, createdBy, institution, category, description, status, priority, assignedTo, attachmentUrl, createdAt, resolvedAt, resolutionNotes | Core entity; status transitions are audited.                     |
| Article      | id, title, body, category, tags                                                                                                                   | Knowledge base content; editable by Administrators only.         |
| Notification | id, userId, title, body, read, createdAt                                                                                                          | Optional persisted history of push notifications (FR-NOT-06).    |
| AuditLog     | id, actorId, action, targetType, targetId, timestamp                                                                                              | Append-only; supports FR-SEC-02.                                 |

## **7. Other Requirements**

### **7.1 Legal and Compliance**

Because ticket and budget-related data may be sensitive government information, the Ministry of Finance should submit terms and Conditions so users will accept them before using the app.

### **7.2 Localization**

Version 1.0 shall support English only. The data model and UI text shall be structured to allow future localization without redesign.

## 8. Use Case Diagram

The diagram below illustrates the primary actors and use cases for the Hyperion Mobile Support Application.

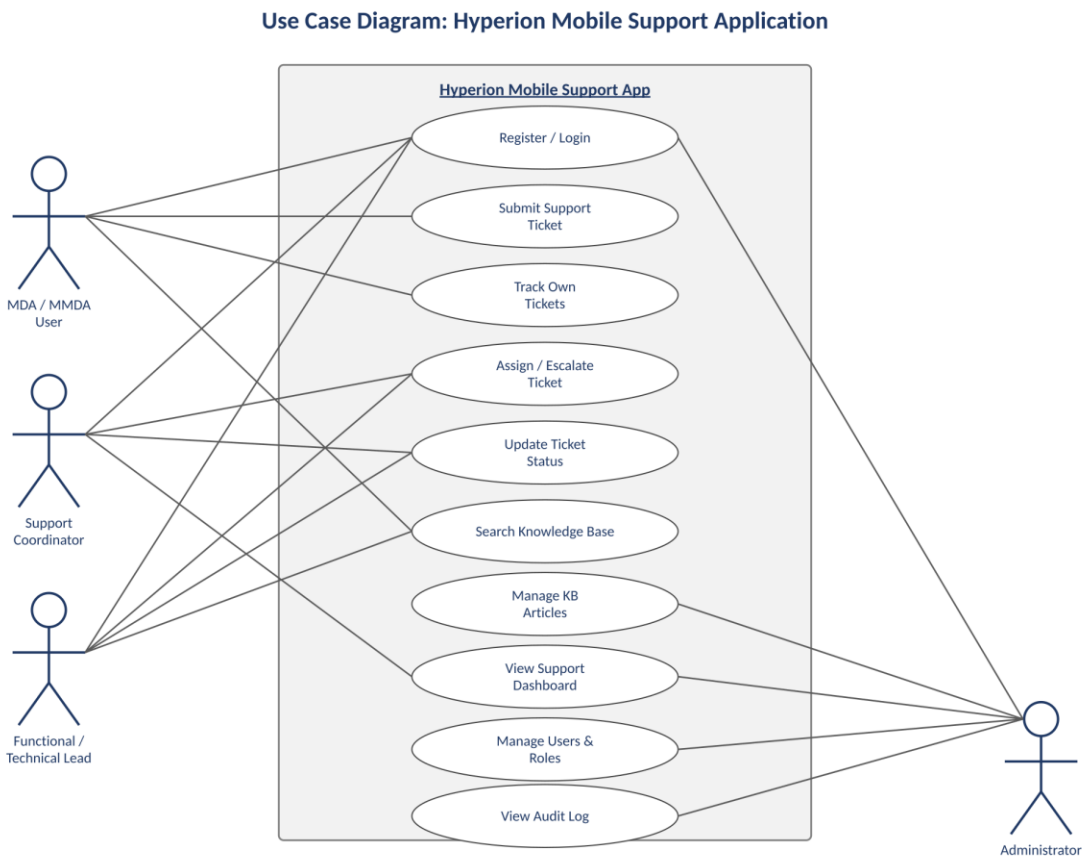


Figure 8.1 — Use case diagram: actors and primary interactions.

## System Flow Diagram

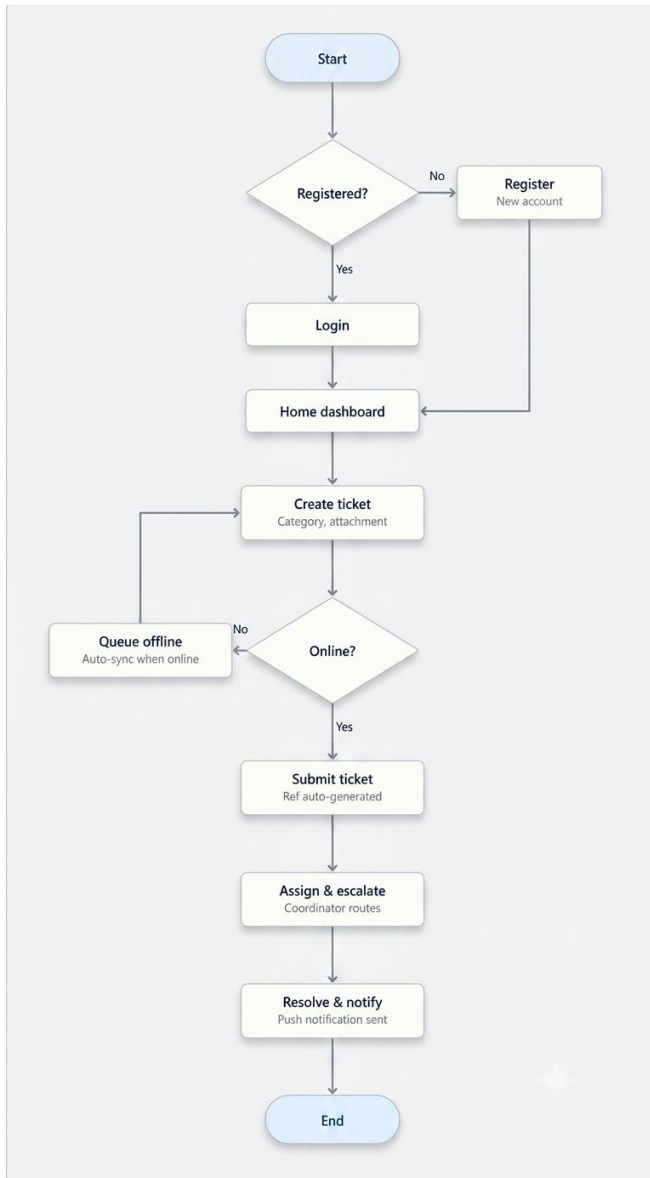


Figure 8.2 — System flow: end-to-end ticket lifecycle.

## Appendix A: Requirements Traceability Matrix

Maps each concept note objective to the requirements that satisfy it, so reviewers can confirm nothing in the source concept note was dropped.

| Concept Note Objective                                       | Related Requirements                                               |
|--------------------------------------------------------------|--------------------------------------------------------------------|
| Objective 1: Improve User Support Across MDAs and MMDAs      | FR-TCK-05, FR-TCK-06, FR-NOT-01, FR-NOT-02, FR-KB-01               |
| Objective 2: Develop a Mobile Support Application            | All functional requirements (Sections 3.1–3.7)                     |
| Objective 3: Improve Data Quality and Budget Cycle Readiness | FR-TCK-02, FR-TCK-12, FR-DSH-01, FR-DSH-02, FR-DSH-04              |
| Objective 4: Build Sustainable Internal Capacity             | FR-KB-01, FR-KB-02, FR-KB-03                                       |
| Data Privacy and Access Control (Risk Mitigation)            | FR-SEC-01, FR-SEC-02, FR-SEC-03, FR-SEC-04, NFR-SEC-01, NFR-SEC-02 |
| Offline and Low-Bandwidth Support                            | FR-OFF-01 through FR-OFF-05, NFR-OFF-01                            |

## **Appendix B: Open Assumptions and Items for Stakeholder Confirmation**

- Final decision on backend hosting (Firebase vs. self-hosted) pending Ministry IT/security review.
- Exact list of institutions (MDAs/MMDAs) to onboard in the pilot is not yet defined.
- Service-level targets in Section 5 (e.g., 99.5% uptime) are draft values pending Ministry agreement.
- Integration with Oracle Hyperion/Essbase APIs, if ever required, is explicitly out of scope for this version and would require a separate SRS addendum.